

Mohamed Elsayed

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Research Objective.....

Developing scalable, general-purpose mechanisms for learning from continuous streams of experience, with a focus on streaming learning, reinforcement learning, and continual learning.

Education

Doctor of Philosophy

University of Alberta, Canada

Computing Science. Advisor: A. Rupam Mahmood

GPA: 4.0/4.0

2022–Present

Master of Science

University of Alberta, Canada

Computing Science. Advisor: A. Rupam Mahmood

GPA: 3.9/4.0

2019–2022

Bachelor of Science

University of Science and Technology, Zewail City, Egypt

GPA: 3.7/4.0

2014–2019

Publications

Conferences and Journals:

1. De Asis, K., **Elsayed, M.**, He, J. (2026). Extending Differential Temporal Difference Methods for Episodic Problems. *Reinforcement Learning Journal* (RLC).
2. Sharifnassab, A., **Elsayed, M.**, De Asis, K., Mahmood, A. R. & Sutton, R. S. (2026). Intentional Updates for Streaming Reinforcement Learning. *International Conference on Machine Learning (ICML)*.
3. Vasan, G., **Elsayed, M.**, Azimi, S. A., He, J., Shahriar, F., Bellinger, C., White, M., & Mahmood, A. R. (2024). Deep policy gradient methods without batch updates, target networks, or replay buffers. *Neural Information Processing Systems (NeurIPS)*.
4. **Elsayed, M.**, Lan, Q., Lyle, C., & Mahmood, A. R. (2024). Weight clipping for deep continual and reinforcement learning. *Reinforcement Learning Journal*, 5(1), 2198–2217 (RLC).
5. **Elsayed, M.**, Farrahi, H., Dangel, F., & Mahmood, A. R. (2024). Revisiting scalable Hessian diagonal approximations for applications in reinforcement learning. *International Conference on Machine Learning (ICML)*.
6. **Elsayed, M.**, & Mahmood, A. R. (2024). Addressing loss of plasticity and catastrophic forgetting in continual learning. *International Conference on Learning Representations (ICLR)*.
7. Zhou, M., Luo, J., Vilella, J., Yang, Y., Rusu, D., Miao, J., ..., **Elsayed M.**, ..., & Wang, J. (2021). SMARTS: Scalable multi-agent reinforcement learning training school for autonomous driving. *Conference on robot learning (CoRL)*.
8. Elgammal, M. A., Elkhoully, O. A., Elhosary, H., **Elsayed, M.**, Mohieldin, A. N., Salama, K. N., & Mostafa, H. (2018). Linear and nonlinear feature extraction for neural seizure detection. *International Midwest Symposium on Circuits and Systems (MWSCAS)*.

Pre-prints and Workshops:

9. **Elsayed, M.**, & Mahmood, A. R. (2025). Multi-stream Sequence Learning. *ICML Workshop on Efficient Systems for Foundation Models*.
10. **Elsayed, M.**, Vasan, G., & Mahmood, A. R. (2024). Streaming deep reinforcement learning finally works. *arXiv preprint arXiv:2410.14606*.
11. **Elsayed, M.**, Vasan, G., & Mahmood, A. R. (2024). Deep reinforcement learning without experience replay, target networks, or batch updates. *NeurIPS Workshop on Fine-Tuning in Modern Machine Learning*.
12. **Elsayed, M.**, & Mahmood, A. R. (2022). Utility-based perturbed gradient descent: An optimizer for continual learning. *NeurIPS Workshop on Optimization for Machine Learning*.
13. **Elsayed, M.**, & Mahmood, A. R. (2022). HesScale: Scalable computation of Hessian diagonals. *NeurIPS Workshop on Higher-Order Optimization in Machine Learning*.
14. **Elsayed, M.**, Hassanzadeh, K., Nguyen, N. M., Alban, M., Zhu, X., Graves, D., & Luo, J. (2020). ULTRA: A reinforcement learning generalization benchmark for autonomous driving. *NeurIPS Workshop on Machine Learning for Autonomous Driving*.

Work Experience

Google DeepMind

Student Researcher

- Working on deep reinforcement learning.

Montreal, Canada

April 2026 - Present

Huawei

Research Intern

- Worked on developing reinforcement learning algorithms for LLM-based agents
- Worked on applying deep reinforcement learning to self-driving cars.

Edmonton, Canada

Dec 2025 - April 2026

May 2020 - Dec 2020

Selected Invited Talks

How Streaming Deep RL unlocks new learning Capabilities

Openmind Winter School

Jan 2026

Kuala Lumpur, Malaysia

The Untapped Potential of Streaming Deep RL Agents

Huawei Canada Research Institute

Aug 2025

Edmonton, Canada

Streaming Deep Reinforcement Learning Finally Works

Google DeepMind Vision Reading Group

Jan 2025

Online

The Potential of Streaming Learning Agents

Scalable Next Generation AI for the Real World Retreat

Dec 2024

Sentosa, Singapore

Streaming Deep Reinforcement Learning Finally Works

Openmind Research Institute Retreat

Oct 2024

Banff, Canada

Utility-based Perturbed Gradient Descent

Lifelong Reinforcement Learning Workshop

February 2023

Holetown, Barbados

Teaching

University of Alberta

Graduate Teaching Assistant

- CMPUT 174: Introduction to the Foundations of Computation
- CMPUT 365: Introduction to Reinforcement Learning
- CMPUT 340: Introduction to Numerical Methods

Edmonton, Canada

Sep 2019 - April 2020

Sep 2022 - Dec 2022

Jan 2024 - April 2024

Zewail City of Science and Technology

Junior Teaching Assistant

- CIE 417: Machine Learning

Cairo, Egypt

Sep 2018 - Dec 2018

Service

- Reviewer** at: ICLR 2023, CoLLAs 2024, RLC 2024, NeurIPS 2024 (Top Reviewer), AISTATS 2025, ICLR 2025, ICML 2025, RLC 2025, TMLR, NeurIPS 2025, AAAI 2025, ICLR 2026, ICML 2026, NeurIPS 2026, and EWRL 2026.
- Evaluator**: Evaluating applicants for the European Lab for Learning & Intelligent Systems (ELLIS) PhD program in 2024 and 2025.